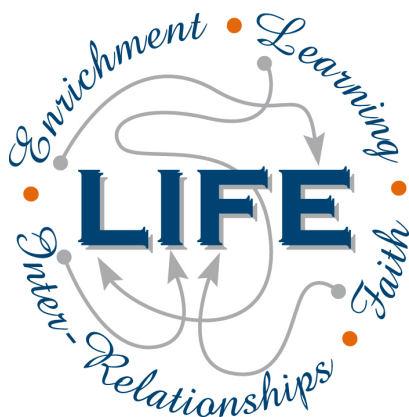




Holy Cross College

Semester 1 Examination 2020

Multiple Choice Booklet

**Year 10 Science
(General)**

Please place your student identification label in this box

Student Name

Student's Teacher

Time allowed for this paper

Reading time before commencing work: 10 minutes
Working time for paper: 100 minutes

Materials required/recommended for this paper

To be provided by the supervisor

This Question/Answer Booklet
Data Sheet

To be provided by the candidate

Standard items: pens, pencils, eraser, correction fluid, ruler, highlighters.

Special items: non-programmable calculators satisfying the conditions set by the School Curriculum and Standards Authority for this course.

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

SECTION 1 MULTIPLE CHOICE (50 marks)

Make a choice of answer and mark it with a cross on the answer sheet in the answer booklet.

1. The alternative forms of each gene are called:
 - (a) alleles.
 - (b) gametes.
 - (c) hybrids.
 - (d) monohybrids.

2. In pea plants, the gene for tall stems (T) is dominant over the gene for short stems (t). The **phenotype** for a pea with the gene combination Tt is:
 - (a) tall stems.
 - (b) middle-size stems.
 - (c) short stems.
 - (d) TT.

3. When red flowered snapdragons are crossed with white flowered snapdragons, some of the offspring have pink flowers. This is an example of:
 - (a) cloning.
 - (b) incomplete dominance.
 - (c) co-dominance.
 - (d) mutation.

4. Which process causes changes in chromosomes to occur?
 - (a) Cell division.
 - (b) DNA mutations.
 - (c) Gene replication.
 - (d) Reproduction.

5. In humans, long eyelashes (L) are dominant over short eyelashes (l). A woman with long eyelashes marries a man with short eyelashes. Two of their four children have short eyelashes. The woman's genotype must be:
 - (a) LL.
 - (b) Ll.
 - (c) ll.
 - (d) long eyelashes.

The next two questions relate to the information below.

The following punnett square shows the likely result of breeding two heterozygous black guinea pigs. The guinea pigs black colour gene (B) dominates the brown colour gene (b).

♀ ♂	B	b
B	BB	Bb
b	Bb	?

6. The **genotype** of the offspring in the incomplete square is:
 - (a) bb.
 - (b) Bb.
 - (c) BB.
 - (d) brown.
7. The **phenotype** of the offspring in the incomplete square would be:
 - (a) black.
 - (b) black and brown.
 - (c) black and white.
 - (d) brown.
8. Dogs have 28 chromosomes in their skin cells. How many chromosomes does a **haploid** dog cell contain?
 - (a) 14
 - (b) 28
 - (c) 1
 - (d) 56
9. Long tails are dominant over short tails in mice, and two hybrid mice are crossed. What percentage of the offspring in the F1 generation should have long tails?
 - (a) 25 %
 - (b) 75 %
 - (c) 0 %
 - (d) 50 %
10. Inheritance of coat colour in shorthorn cattle is an example of **co-dominance**. The genotype RR produces a red coat, WW a white coat and RW a roan coloured coat. If a roan cow is crossed with a red bull, what percentage of the offspring would be expected to be red?
 - (a) 50%
 - (b) 75%
 - (c) 25%
 - (d) 100%

11. DNA stands for:
- (a) dinucleotide acid.
 - (b) ribonucleic acid.
 - (c) deoxyribonucleic acid.
 - (d) dioxyriboncleus acid.
12. _____ contain half the original number of chromosomes and are said to be haploid.
- (a) Zygotes
 - (b) Gametes
 - (c) Ganglia
 - (d) Gonads
13. Mitosis results in:
- (a) four identical cells with half the number of chromosomes as the parent cell.
 - (b) four identical cells with the same number of chromosomes as the parent cell.
 - (c) two identical cells with the same number of chromosomes as the parent cell.
 - (d) two identical cells with half the number of chromosomes as the parent cell.
14. Dominant genes mask the presence of _____ genes.
- (a) heterozygous
 - (b) pure-bred
 - (c) recessive
 - (d) homozygous
15. In a pedigree chart, how is a female who shows a specific trait represented?
- (a) An unshaded square.
 - (b) An unshaded circle.
 - (c) A shaded square.
 - (d) A shaded circle.
16. How many **autosomal** chromosomes do humans have?
- (a) 46
 - (b) 23
 - (c) 22
 - (d) 44
17. The primary purpose of meiosis is:
- (a) to produce new skin cells.
 - (b) to produce gametes.
 - (c) for growth and repair.
 - (d) for DNA replication.

18. A dominant trait is one that is expressed in a:
- (a) heterozygous individual where they have two of the same alleles, and that allele is expressed.
 - (b) heterozygous individual where they have two different alleles, but only one allele is expressed.
 - (c) homozygous individual where they have two different alleles, but only one allele is expressed.
 - (d) homozygous individual where they have two of the same alleles, and that allele is expressed.
19. Let us say that the gene that codes for normal-type wings in fruit flies has the symbol 'N' and the gene that codes for vestigial (stunted) wings has the symbol 'n'. The genotype of heterozygous flies would be:
- (a) Nn.
 - (b) normal wings.
 - (c) NN.
 - (d) nn.
20. In a cross between a pure-bred dominant individual (for a gap in the front teeth) and a pure-bred recessive individual, the chance of a hybrid child is:
- (a) 25%.
 - (b) 50%.
 - (c) 75%.
 - (d) 100%.
21. If a sperm with an X chromosome fertilises an egg, what sex will the baby be?
- (a) It is difficult to tell from the information given.
 - (b) It has a 50% chance of being male or female.
 - (c) Female.
 - (d) Male.
22. Where in the cell are chromosomes located?
- (a) In the cytosol.
 - (b) In the nucleus.
 - (c) In the vacuoles.
 - (d) In the cell membrane.
23. What is a black hole?
- (a) A rapidly spinning star that gives out neutrons.
 - (b) A dark invisible hole in space that is a worm hole into another parallel universe.
 - (c) A massive explosion that gives out huge amounts of light.
 - (d) The remains of a star that is so massive that light can not escape from it.
24. What are galaxies? Select the best answer from the options below.
- (a) Huge masses of gas that explode.
 - (b) Collections of stars and solar systems and gas clouds.
 - (c) Shining objects in the sky.
 - (d) A star, along with its planets and their moons.

25. A star like our Sun has a life cycle. Each of the lists below contains events from this life cycle. Which list has the events in the correct order?
- (a) Nebula, protostar, star, red giant, planetary nebula.
 - (b) Protostar, planetary nebula, nebula, star, red giant.
 - (c) Protostar, nebula, red giant, star, planetary nebula.
 - (d) Nebula, planetary nebula, red giant, protostar, star.
26. If the light from a star is blue shifted, that tells us that:
- (a) the star must be moving towards us.
 - (b) the star must be moving away from us.
 - (c) the star must be moving parallel to us.
 - (d) the star must have a solar system of planets.
27. The Universe is expanding. This is usually taken to be evidence that:
- (a) the universe began in a Big Bang.
 - (b) we are at the centre of the Universe.
 - (c) gravity only works near the Earth.
 - (d) the universe was created by a Supreme Being.
28. A light year is defined as:
- (a) the distance light travels from the closest star.
 - (b) the distance from Alpha Centauri to the sun.
 - (c) the distance of the Earth from the sun.
 - (d) the distance light travels in a year.
29. The main fuel of stars is:
- (a) hydrogen.
 - (b) helium.
 - (c) carbon.
 - (d) gold.
30. A star ten times the size of our sun would go through the following stages:
- (a) supernova, neutron star, pulsar, black hole.
 - (b) pulsar, neutron star, black hole, supernova.
 - (c) neutron star, pulsar, supernova, black hole.
 - (d) black hole, supernova, pulsar, neutron star.
31. A galaxy is a group of many billions of stars held together by _____.
- (a) gravitational forces.
 - (b) black holes.
 - (c) magnetic forces.
 - (d) quasars.
32. The Milky Way is most accurately described as a _____.
- (a) planet.
 - (b) universe.
 - (c) solar system.
 - (d) galaxy.

33. During the nuclear fusion reactions occurring in the Sun's core:
- (a) hydrogen particles join to form helium particles, and energy is absorbed.
 - (b) helium particles join to form hydrogen particles, and energy is released.
 - (c) hydrogen particles join to form helium particles, and energy is released.
 - (d) helium particles join to form hydrogen particles, and energy is absorbed.
34. In which galaxy is our solar system located?
- (a) Andromeda
 - (b) Sombrero
 - (c) Milky Way
 - (d) Whirlpool
35. What type of reaction fuels the vast amount of energy produced by a star?
- (a) Nuclear fission
 - (b) Chemical
 - (c) Physical
 - (d) Nuclear fusion
36. When was the Big Bang theory proposed?
- (a) 1948 by George Gamow and Ralph Alpher.
 - (b) 1933 by Albert Einstein.
 - (c) 1927 by Fred Hoyle.
 - (d) 1927 by Georges Lemaitre.
37. Which of the following is the best definition of a constellation?
- (a) A grouping of planets in an apparent pattern.
 - (b) Patterns of stars that change.
 - (c) Stars that are in fixed positions together.
 - (d) A grouping of stars in an apparent pattern.
38. To date, approximately how far away is the most distant object in the known universe?
- (a) 61 billion light years
 - (b) 10 billion light years
 - (c) 51 billion light years
 - (d) 15 billion light years
39. What is the key difference between the formation of a **supernova** and the formation of a **white dwarf**?
- (a) The starting mass of the star.
 - (b) The position in the galaxy.
 - (c) If there are planets in the system.
 - (d) Whether the star was hit by an asteroid.
40. How long ago was the Big Bang?
- (a) 15 billion years ago
 - (b) 10 trillion years ago
 - (c) 1 billion years ago
 - (d) 65 million years ago

41. Of what are nebulas made?
- (a) Asteroid belts.
 - (b) Stellar clouds of gas and dusts.
 - (c) Stellar masses of gas and energy.
 - (d) Planets and their magnetic fields.
42. What does the Steady State theory propose?
- (a) The universe and the galaxies in it stay in the same place.
 - (b) The universe has always existed.
 - (c) The universe remained at the same size after it formed.
 - (d) The universe maintains the same number of galaxies.
43. Which of the following is the best definition of a star?
- (a) A sphere of mostly hydrogen and helium gas that shines brightly and is held together by gravity.
 - (b) A sphere of mostly helium gas that shines brightly and forms gravity.
 - (c) A sphere of solid hydrogen that shines brightly and is held together by gravity.
 - (d) A sphere of gas that shines brightly and forms gravity.
44. What does the equation $E = mc^2$ describe?
- (a) The conversion of matter into energy.
 - (b) The conversion of matter into entropy.
 - (c) The conversion of light into heat energy.
 - (d) The conversion of energy into gravity.
45. What is **climate**?
- (a) A description of the atmosphere in terms of wind, precipitation and temperature.
 - (b) A description of the hydrosphere in terms of tide, acidity and temperature.
 - (c) An average of the weather conditions over a short period of time.
 - (d) An average of the weather conditions over a long period of time.
46. The biosphere contains:
- (a) All of the planet's animals.
 - (b) All of the planet's plants
 - (c) All of the planet's living organisms.
 - (d) The Earth's crust and soils.
47. How is **climate change** causing sea levels to rise?
- (a) Decreasing sunlight is evaporating less water.
 - (b) Increasing temperatures are melting the ice caps.
 - (c) Decreasing gravity is pulling on the water less.
 - (d) Increasing rainfall is filling up the oceans.
48. In relation to the carbon cycle, describe what photosynthesis allows plants to do.
- (a) Absorb methane from Earth's atmosphere.
 - (b) Absorb O_2 from Earth's atmosphere.
 - (c) Absorb CO_2 from Earth's atmosphere.
 - (d) Absorb H_2O from Earth's atmosphere.

49. Methane and CO₂ are **greenhouse gases**. What does this mean?
- (a) They let heat escape through the atmosphere and decrease Earth's temperature.
 - (b) They speed up the growth of plants.
 - (c) They trap heat in the atmosphere and increase Earth's temperature?
 - (d) They are a type of fertiliser.
50. How does phosphorus move from rocks to the soil?
- (a) Osmosis and weathering.
 - (b) Rain and weathering.
 - (c) Osmosis and rain.
 - (d) Teleportation and weathering.

END OF SECTION 1